Institution *KU Leuven*
Position *Postdoctoral researcher*
Website *https://diego-mmg.github.io/*
Email *diego.martinez@kuleuven.be*

Diego Martínez (cv)

Work Experience

FWO FWO senior postdoctoral researcher **University:** *KU Leuven*, *Department of Mathematics*
Supervisor: Prof. Gábor Szabó from October 2025 until now

Postdoc Postdoctoral researcher **University:** *KU Leuven*, *Department of Mathematics*
Supervisor: Prof. Gábor Szabó from September 2024 until September 2025

Postdoc Postdoctoral researcher **University:** *University of Münster*, *Mathematical Institute*
Supervisor: Prof. Wilhelm Winter from October 2021 until August 2024

PhD Ph.D. student **Institutions:** *Universidad Carlos III de Madrid* and *Instituto de Ciencias Matemáticas*
Supervisor: Prof. Fernando Lledó from April 2017 until March 2021

Master Master's student **University:** *Universidad Autónoma de Madrid*, *Departamento de Matemáticas*
Supervisor: Prof. Fernando Lledó from September 2016 to August 2017

Education

Ph.D. University: *Universidad Carlos III de Madrid* (UC3M) and *Instituto de Ciencias Matemáticas* (ICMAT)
Supervisor: *Prof. Fernando Lledó*, Program: *Ph.D. in Mathematics* (international distinction)

Thesis: *Amenability and coarse geometry of (inverse) semigroups and C*-algebras*

Grade: *excellent - cum laude* from September 2016 to March 2021

Master University: *Universidad Autónoma de Madrid* (UAM)
Supervisor: *Fernando Lledó*, Program: *Master Program in Mathematics and Applications* (60 ects)

Thesis: *Amenability, paradoxical decompositions and applications*

Grade: *10/10* from September 2015 to July 2016

Degree University: *Universidad Autónoma de Madrid* (UAM) from September 2010 to June 2015
Program: *Graduate program in Computer Engineering*

Degree University: *Universidad Autónoma de Madrid* (UAM) from September 2010 to June 2015
Program: *Graduate program in Mathematics*

Thesis: *Entropía y códigos de Huffman* (in Spanish, **grade** 10/10)

Publications

[15] D. Keuppens and D. Martínez, *A groupoid model for Rørdam's finite-infinite C*-algebra*, https://arxiv.org/pdf/2610.01543 (2026).

[14] D. Martínez, *A Baum-Connes assembly map for essential semigroup crossed products*, https://arxiv.org/abs/2606.24502 (2026).

[13] D. Martínez and F. Vigolo, *A rigidity framework for Roe-like algebras*, accepted for publication in Mem. Amer. Math. Soc. (2026).

[12] Becky Armstrong, Lisa Orloff Clark, Astrid an Huef, D. Martínez and I. Tolich, *A Dichotomy for Inverse–Semigroup Crossed Products via Dynamical Cuntz Semigroups*, accepted for publication in Münster J. Math. (2026).

[11] D. Martínez and N. Szakács, *Algebraic singular functions are not always dense in the ideal of C^*-singular functions*, Bull. London Math. Soc. (2025) DOI: 10.1112/blms.70381.

[10] A. Buss and D. Martínez, *Essential groupoid amenability and nuclearity of groupoid C*-algebras*, J. Funct. Anal. **290** (2026) 111427 (pp. 54) DOI: 10.1016/j.jfa.2026.111427.

[9] D. Martínez and F. Vigolo, *Roe algebras of coarse spaces via coarse geometric modules*, JNCG (2025) DOI: 10.4171/JNCG/642.

[8] D. Martínez and F. Vigolo, *C*-rigidity of bounded geometry metric spaces*, Publ. Math. IHES (2025) DOI: 10.1007/s10240-025-00155-3.

[7] Y. C. Chung, D. Martínez and N. Szakács, *Quasi-countable inverse semigroups as metric spaces, and the uniform Roe algebras of locally finite inverse semigroups*, to appear in Groups, Geom. Dyn. (2025).

[6] A. Buss and D. Martínez, *Approximation properties of Fell bundles over inverse semigroups and non-Hausdorff groupoids*, Adv. Math. **431** (2023) 109251 (pp. 54) DOI: 10.1016/j.aim.2023.109251.

[5] D. Martínez, *A note on the quasi-diagonality of inverse semigroup reduced C*-algebras*, J. Oper. Th. **90** (2023) (pp. 17) DOI: 10.7900/jot.2021nov29.2424.

[4] F. Lledó and D. Martínez, *A note on commutation relations and finite dimensional approximations*, Expo. Math. (2022) DOI: 10.1016/j.exmath.2022.08.004.

[3] F. Lledó and D. Martínez, *The uniform Roe algebra of an inverse semigroup*, J. Math. Anal. Appl. **499** (2021) 124996 (pp. 28) DOI: 10.1016/j.jmaa.2021.124996.

[2] P. Ara, F. Lledó and D. Martínez, *Amenability and paradoxicality in semigroups and C*-algebras*, J. Funct. Anal. **279** (2020) 108530 (pp. 43) DOI: 10.1016/j.jfa.2020.108530.

[1] F. Lledó and D. Martínez, *Notions of infinity in quantum physics*, 60 Years Alberto Ibort Fest Classical and Quantum Physics, Chapter 14, Springer (2019) (pp. 14) DOI: 10.1007/978-3-030-24748-5_14.

Other articles

[2] D. Martínez, *On the relation between the product of KK-groups and the KK-group of the product*, https://arxiv.org/abs/2606.20181 (2026).

[1] D. Martínez, *Amenability notions and Fell bundles*, extended abstract, Oberwolfach Rep. 21 (2024), 761–764 DOI.

Selected International Talks

- *A groupoid model for Rørdam's C^*-algebra with an infinite and a finite projection*,
  6th Int. Conf. Settat on Operator Alg. and Appl. – ICSOAA2026, Settat, Morocco — Nov. 2026
- *On induced local symmetries of the Cuntz semigroup arising from actions*,
  Workshop on Cuntz semigroups and dynamics, Workshop, Gothenburg, Sweden — Sept. 2026
- *A Gelfand-type duality and C*-rigidity for metric spaces*,
  Groups, dynamics and operator algebras, Conference, Glasgow, United Kingdom — Aug. 2026
- *Rigidity and outer automorphisms of Roe algebras*,
  Operator Algebras and Applications, Mini-Workshop, Erlangen, Germany — July 2025
- *Finite distortion operators, Roe algebras, and coarse invariants*,
  Functional and Metric Analysis, and their Interactions, BIRS, Spain — May 2025
- *Cartan subalgebras, cp maps, and their inductive limits*,
  Cartan Subalgebras in Operator Algebras, and Topological Full Groups, BIRS, Canada — Nov. 2024

- *Amenability of essential inverse semigroup C*-algebras*,
  Plenary talk, SEmigroups, grOUpoids and C*-aLgebras, KIAS, Korea — July 2024
- *Roe algebras are rigid*,
  Analysis Seminar, Newcastle University, UK — May 2024
- *Amenability notions and Fell bundles*,
  Combinatorial *-algebras, MFO, Oberwolfach — March 2024
- *Large-scale geometry of group-like objects*,
  Colloquium, University of Göttingen — May 2023
- *Minicourse on C*-algebras, their diagonals, and (inverse) semigroup actions*,
  Topics in Geometric Semigroup Theory, University of York — June 2022
- *Exactness and geometric properties of inverse semigroups*,
  Non-commutativity in the North, Göteborgs Universitet — March 2022
- *C* vs coarse properties of inverse semigroups*,
  Analysis online seminar, University of Glasgow — Oct. 2021
- *Schützenberger graphs and property A, or how to see exactness of the reduced semigroup C*-algebra*,
  C*-algebras and geometry of groups and semigroups, University of Oslo — March 2021
- *Non-group like metrics appearing in group-like objects*, — Dec. 2020
  UK Virtual Operator Algebras Seminar, United Kingdom
- *Some quasi-isometric invariants for inverse semigroups*, — May 2020
  York semigroup (online talk), University of York, United Kingdom
- *Coarse geometry and inverse semigroups*, — Nov. 2019
  Colloquium talk, University of Montana, USA
- *Traces in inverse semigroup Roe algebras*, — Aug. 2019
  YMC*A, University of Copenhagen, Denmark
- *Amenability in inverse semigroups*, — June 2019
  SandGAL-2019, University of Cremona, Italy
- *Amenability in inverse semigroups and C*-algebras*, — Dec. 2018
  II Joint Meeting Spain-Brazil in Mathematics, Universidad de Cádiz, Spain

Teaching

- **Functional Analysis**, at KU Leuven, — Fall 26
- **Master's Thesis**, at KU Leuven, — Fall 25/Spring 26
- **Master Thesis Reader**, at KU Leuven, — Spring 2025
- **Bachelor's Final Degree Project**, at KU Leuven, — Spring 2026
- **Functional Analysis**, at KU Leuven, — Fall 25
- **Bachelor's Final Degree Project**, at KU Leuven, — Spring 2025
- **Master Thesis Reader**, at KU Leuven, — Spring 2025
- **Student Teaching Assistant**, at KU Leuven, — Fall 2024
  Differentiaalvergelijkingen (differential equations), Reeks 11+12. Bachelor in Bio-engineering.
- **Cartan Subalgebras**, at the University of Münster — Fall 2022
  Reading seminar
- **TFG committee member**, at UC3M — Fall 2020
- **Quasi-diagonality and the Tikuisis–White–Winter Theorem**, at Texas A&M — Fall 2019
  Reading seminar
- **TFG committee member**, at UC3M — Spring 2019

- **Student Teaching Assistant**, at UC3M Fall 2018
  Cálculo I (calculus I), group 45. Bachelor in Aerospace Engineering
- **Student Teaching Assistant**, at UC3M Fall 2017
  Álgebra lineal I (linear Algebra I), group 49. Bachelor in Energy Engineering

Participation in committees

- Member of the **Party Committee**, at KU Leuven 2024 – present.
- Member of the **Early Career Committee**, at the University of Münster 2023 – 2024.

Reviewer of

- zbMATH Open 2022 – present.
- MathSciNet 2023 – present.

Scholarships and Awards

- **FWO Senior Postdoctoral Researcher** (FWO) - Research Foundation, Flanders 2025 – now
  Grant for senior postdoctoral researchers in the Flanders region, Belgium. Allows to conduct research as one sees fit. Funds all of the researcher's salary and travel funds.
- **Formación de personal investigador** (FPI) - Severo Ochoa 2017 – 2021
  Grant supporting the fulfillment of the PhD from the MINECO (Spanish Ministry of Economy)
- **Mathematics Department Scholarship** (UAM) 2015 – 2016
  Grant intended to support the study of the Master in Mathematics (one of the top 5)
- **BIT Summer School in Beijing** 2012
  Grant awarded to pay for an intensive course on Chinese culture and language

Research Visits

- **University of Wollongong**, Australia Oct.–Nov. 2023
- **Texas A&M**, USA Aug.–Dec. 2019
  Department of Mathematics
- **UAB**, Spain 2017 & 18
  Centre de Recerca Matematica
- **University of Münster**, Germany May 2016
  Institute of Theoretical Mathematics

Organizer of

- **Junior Seminar**, ICMAT Sep. 2018–June 2021
  Local seminar for Ph.D. students
- **OAGAQI** thematic semester, March–June 2019
  Thematic semester about operator algebras, groups and applications to quantum information
- **JAE School**, ICMAT June 2017
  School that serves as an introduction to research for undergraduate students
- **Simon Marais Math Competition**, KU Leuven Fall 2026

Outreach

- **Outreach newspaper** article in *El País* 2019
  – *Ganar juegos usando la mecánica cuántica*
- **Outreach newspaper** article in *El País* 2017
  – *Dándole vueltas al infinito*

Languages

- **Spanish:** Native
- **English:** Proficient (C2)
- **German:** Beginner (A2)
- **Dutch:** Beginner (A1)